Math 280, Quiz #5 **Name**_____

version A

You must show your work and/or justify your answers for full credit.

1. (2 pts each) Determine whether the following improper integrals converge or diverge. If they converge, give the val they converge to.

a. $\int_2^\infty \frac{dx}{x \ln x}$

b. $\int_0^\infty e^{-2x} dx$

2. (2 pts) Evaluate the geometric series $\frac{1}{3} + \frac{2}{15} + \frac{4}{75} + \frac{8}{375} + \ldots$

3. (2 pts each) Find a closed form for the following infinite series.

a. $x + 2x^3 + 3x^5 + 4x^7 + \ldots$

b. $e^x + \frac{e^{2x}}{2} + \frac{e^{3x}}{3} + \frac{e^{4x}}{4} + \ldots$

Math 280, Quiz #5 **Name**_____

version B

You must show your work and/or justify your answers for full credit.

1. (2 pts each) Determine whether the following improper integrals converge or diverge. If they converge, give the val they converge to.

a. $\int_0^\infty e^{-3x}dx$

b. $\int_2^\infty \frac{dx}{x\ln x}$

2. (2 pts) Evaluate the geometric series $\frac{1}{2}+\frac{3}{10}+\frac{9}{50}+\frac{27}{250}+...$

3. (2 pts each) Find a closed form for the following infinite series.

a. $e^x+\frac{e^{2x}}{2}+\frac{e^{3x}}{3}+\frac{e^{4x}}{4}+...$

b. $x+2x^3+3x^5+4x^7+...$

Math 280, Quiz #6 **Name**_____

version A

You must show your work and/or justify your answers for full credit.

1. (2 pts) Use the comparison test to determine whether the series $\sum_{n=1}^{\infty} \frac{\cos^2 n}{n^{3/2}}$ converges or diverges.

2. (2 pts) Use the integral test to determine whether the series $\sum_{n=2}^{\infty} \frac{1}{n\sqrt{\ln n}}$ converges or diverges.

3. (2 pts each) Determine whether the following series converge or diverge. Clearly state which test you are using.

a. $\sum_{n=1}^{\infty} \frac{n+2}{n^3+2n+5}$

b. $\sum_{n=1}^{\infty} \frac{n+1}{\sqrt{n^3+4}}$

c. $\sum_{n=1}^{\infty} \frac{5^n}{(2n)!}$

Math 280, Quiz #6 **Name**_____

version B

You must show your work and/or justify your answers for full credit.

1. (2 pts) Use the integral test to determine whether the series $\sum_{n=2}^{\infty} \frac{1}{n\sqrt{\ln n}}$ converges or diverges.

2. (2 pts) Use the comparison test to determine whether the series $\sum_{n=1}^{\infty} \frac{\cos^2 n}{n^{3/2}}$ converges or diverges.

3. (2 pts each) Determine whether the following series converge or diverge. Clearly state which test you are using.

a. $\sum_{n=1}^{\infty} \frac{n+3}{n^3+5n+2}$

b. $\sum_{n=1}^{\infty} \frac{n+2}{\sqrt{n^3+3}}$

c. $\sum_{n=1}^{\infty} \frac{3^n}{(2n)!}$

Math 280, Quiz #7 **Name**_____

version A

You must show your work and/or justify your answers for full credit.

1. (2 pts) Draw two pictures to explain how you obtain upper and lower bounds for $\sum_{n=20}^{\infty} \frac{1}{n^3}$. Please give numerical answers (e.g., $5/11^3$), but do not give a decimal approximation.

2. a. (1 pt) Given that $\arctan\left(\frac{1}{n^2+n+1}\right) = \arctan\left(\frac{1}{n}\right) - \arctan\left(\frac{1}{n+1}\right)$, simplify $S_N = \sum_{n=1}^{N} \arctan\left(\frac{1}{n^2+n+1}\right)$.

b. (1 pt) Evaluate $\sum_{n=1}^{\infty} \arctan\left(\frac{1}{n^2+n+1}\right)$

3. (2 pts each) Determine whether the following series converge absolutely, converge conditionally, or diverge. Clearly state which test you are using.

a. $\sum_{n=1}^{\infty} \frac{(-3)^n}{n!}$

b. $\sum_{n=1}^{\infty} (-1)^n \frac{n+1}{n^2+1}$

c. $\sum_{n=1}^{\infty} (-1)^n \cos\left(\frac{1}{n}\right)$

Math 280, Quiz #7 **Name**_____

version B

You must show your work and/or justify your answers for full credit.

1. (2 pts) Draw two pictures to explain how you obtain upper and lower bounds for $\sum_{n=30}^{\infty} \frac{1}{n^3}$. Please give numerical answers (e.g., $5/11^3$), but do not give a decimal approximation.

2. a. (1 pt) Given that $\arctan\left(\frac{1}{n^2+n+1}\right) = \arctan\left(\frac{1}{n}\right) - \arctan\left(\frac{1}{n+1}\right)$, simplify $S_N = \sum_{n=1}^{N} \arctan\left(\frac{1}{n^2+n+1}\right)$.

b. (1 pt) Evaluate $\sum_{n=1}^{\infty} \arctan\left(\frac{1}{n^2+n+1}\right)$

3. (2 pts each) Determine whether the following series converge absolutely, converge conditionally, or diverge. Clearly state which test you are using.

a. $\sum_{n=1}^{\infty} (-1)^n \cos\left(\frac{1}{n}\right)$

b. $\sum_{n=1}^{\infty} (-1)^n \frac{n-1}{n^2+1}$

c. $\sum_{n=1}^{\infty} \frac{(-2)^n}{n!}$